**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING**
**NORTH CAROLINA STATE UNIVERSITY**

**CHE 450** **Homework Set 6**

*For Problem 1:*

- *Submit written work to Gradescope link similar to previous HW sets*

**Problem 1 (8%)** Suppose you are planning on producing the gasoline additive methylcyclopentadienyl manganese tricarbonyl (MMT) in a 2500 gallon liquid-phase stirred batch reactor through the reduction of bis(methylcyclopentadienyl) manganese by triethylaluminium; this reaction is highly exothermic ($\Delta H < -150$ kJ/mol). Describe two strategies you would consider incorporating into the process design to safely remove the exothermic heat released in the reactor in order to avoid a runaway reaction.

**ASPEN Problems:**

*For ASPEN problems, submit the following to Moodle:*

- *The saved ASPEN Plus files (file name: ASPEN_HW6_Name_Problem#)*
- *A Word or PDF file (file name: HW6_Name_Combined) containing the following details for each problem:*
    1) *Screen capture of any 'Input' screens*
    2) *Screen capture of 'Stream Summary' section output showing data for all streams*
    3) *Final answer(s) to the problem*
    4) *Any additional screen captures requested in the problem statement*

**Problem 2 (16%)** Model the flash separation of a 1000 kmol/hr stream containing 70 mole% benzene and 30 mole% toluene at 145°C and 4 bar. Use a Flash2 separator and the RK-SOAVE method. What is the mole fraction of benzene in the vapor stream coming off the flash separator?

**Problem 3 (16%)** Model the separation of the same stream as in Problem 1, but instead of feeding the stream to a Flash2 separator, the stream is fed to a DSTWU column with 14 stages and recoveries of the heavy key and light key of 0.03 and 0.97, respectively. Use the RK-SOAVE method and assume the condenser and reboiler each operate at a pressure of 3.8 bar. What is the minimum reflux ratio (shown on the DSTWU Results screen) that can be used for this separation?

**Problem 4 (40%)** You are working at a start-up company aiming to produce ethylene from a 100 kg/hr stream of ethane at conditions of 400°C and 30 psig that is fed to an adiabatic PFR that is 3 m long and 0.25 m in diameter according to the following reversible reaction:

$$C_2H_6 \leftrightarrow C_2H_4 + H_2$$

For the reversible reaction shown above, the forward reaction is known to have kinetic values of k = 1125, n = 2, and E = 10 kJ/mol, while the backward reaction is known to have kinetic values of k = 70000, n = 2, and E = 5000 kJ/mol. Unfortunately, the following undesired side reaction also occurs which consumes some of the ethane fed to the process:

$$C_2H_6 + H_2 \rightarrow 2\ CH_4$$

It is known this reaction has kinetic values of k = 12000, n = 2, and E = 25 kJ/mol. Assume all reactions in this system adhere to a power law model of the type $-r = kT^n e^{-E/RT}$ and use molarity as the basis for concentrations. Considering the information above, and using the PENG-ROB method, use ASPEN's RPlug block to model the PFR. What is the flowrate of methane (in kg/hr) exiting the reactor?

**Problem 5 (20%)** Suppose that in contrast to Problem 4, we do <u>not</u> know the kinetic parameters for the specified reactions; instead, we plan to use Gibbs free energy minimization to determine reactor behavior (assume the species listed in the reactions in Problem 3 are the only species possible in the system, and also assume that the reactor operates at the conditions of the inlet feed to the reactor). Considering the information above, and using the PENG-ROB method, use ASPEN's RGibbs block to model the system. What is the flowrate of methane (in kg/hr) exiting the reactor?